

Customer Retention & Churn Analysis

Project Overview

This project analyzes customer churn patterns using a telecom customer dataset. The objective is to identify factors influencing customer churn and provide recommendations to improve customer retention.

Tools Used

- Python (Pandas, NumPy)
- Tableau Public
- Excel (optional)

Data Cleaning

- Removed 11 records with missing TotalCharges values.
- Converted TotalCharges from object type to numeric.
- Verified dataset quality and consistency.

Key Metrics

Metric	Value
Total Customers	7032
Churned Customers	1869
Retained Customers	5163
Churn Rate	26.58%
Retention Rate	73.42%
Average Monthly Charges	64.8
Average Customer Lifetime	32.42 months
Total Revenue	16,056,168.7

Findings

Contract Type

Month-to-month customers exhibited the highest churn behavior, while customers on one-year and two-year contracts showed significantly better retention.

Internet Service

Fiber optic customers had the highest churn count, suggesting potential service quality or pricing concerns.

Payment Method

Customers using Electronic Check showed the highest churn rates compared to other payment methods.

Senior Citizen Analysis

Senior citizens showed notable churn behavior, indicating a need for targeted customer support and engagement strategies.

Recommendations

1. Encourage customers to move from month-to-month contracts to annual contracts.
2. Investigate service quality and pricing for Fiber Optic customers.
3. Promote automatic payment methods through incentives.
4. Develop targeted retention campaigns for at-risk customer groups.
5. Offer loyalty rewards for long-tenure customers.

Conclusion

The analysis identified contract type, internet service, payment method, and customer demographics as major factors influencing churn. Implementing targeted retention strategies can help reduce customer loss and improve long-term business performance.